

Dynamic mTORC1: your hypothesis just got direct experimental support

A literature check (PubMed + web, 2023–2026) against three open questions in the Atlas, run to see whether they still hold, have closed, or have shifted.

Headline finding — your central idea now has direct evidence behind it.

Your working hypothesis is that a cell's fate (autophagy vs. growth) is decided by the *pattern* of mTORC1 activation, not just its average level. Joshi et al. (*Cell Reports*, 2024) show that mTORC1 activity naturally oscillates across the cell cycle — lowest in mitosis/G1, highest in S/G2 — driven by the TSC complex, independent of Akt or Mek/Erk. Critically, cells with low mTORC1 in G1 (the trough of the oscillation) are *more* sensitive to autophagy induction from the exact same partial inhibition or nutrient drop that a cell in S/G2 (the peak) would shrug off. Two cells, same average mTORC1 signal, different autophagy outcome — because of timing. DOI: 10.1016/j.celrep.2024.114543.

What shifts as a result: the "clock" driving mTORC1 pulses isn't only food and circadian rhythm — it's also the cell cycle itself, via TSC and CDK1. And the tool needed to actually test pulse hypotheses now exists: AIMTOR, a live BRET biosensor that reads mTOR activity in single cells and specific subcellular compartments in real time (Bouquier et al., *BMC Biology*, 2020, DOI: 10.1186/s12915-020-00790-8).

Verdict on each open question

Q1

RECENTLY ADDRESSED

"Can mTORC1 be selectively inhibited without disrupting mTORC2?"

Largely yes, as of 2023–2025. This is the single biggest concrete change in the field since the Atlas was written. Bi-steric molecules — a rapamycin-like arm chemically linked to an active-site inhibitor — now hit mTORC1 with >25× selectivity over mTORC2.

Preclinical (2023): RMC-6272 completely suppressed mTORC1 (hitting the rapamycin-resistant substrate 4E-BP1) and overcame hormone- and CDK4/6-inhibitor resistance in ER+/HER2– breast cancer cell lines and patient-derived xenografts. Meng et al., *Oncogene*. DOI: 10.1038/s41388-023-02737-z.

Clinical (2025): its successor RMC-5552 completed a first-in-human Phase 1 trial (n=57, solid tumors). Treatment-related hyperglycemia was only 4% and not dose-limiting — direct clinical confirmation that sparing mTORC2 avoids the metabolic toxicity that limits chronic rapamycin, while the disease control rate reached 64%. Schram et al., *Clinical Cancer Research*. DOI: 10.1158/1078-0432.CCR-25-2112.

What's still open: whether this selectivity translates into a longevity/healthspan benefit outside oncology — nobody has run the aging-relevant version of this trial yet.

mTORC1 mTORC2 bi-steric MEN2023 SCH2025

Q2

PARTIALLY ADDRESSED

"What is the optimal pulsatile (on/off) dosing schedule for mTOR inhibition?"

Still open, but noticeably more tractable than before. Three things changed the picture: (a) mTORC1 activity is now shown to oscillate naturally through the cell cycle (Joshi 2024); (b) live single-cell biosensors (AIMTOR) exist to measure that on/off pattern directly instead of inferring it; (c) an insulin-mTOR feedback model shows time- and dose-dependent biphasic mTOR behavior driven by mTORC1 → mTORC2 negative feedback. What nobody has measured yet: which specific pattern — frequency, amplitude, or on/off ratio — is optimal for a given outcome. That remains genuinely open territory, and it's the organism-level analogue of intermittent fasting / time-restricted eating.

Rapamycin BIT2016 JOS2024 BOU2020

"Degron-based oscillatory control of mTORC1 in hepatocytes"

Still an open, unclaimed experiment — but the literature refines what "degron" should mean here. Most current mTOR literature uses "degron" for *endogenous* degron pathways that mTORC1 itself regulates (e.g. N-degron/CTLH E3 ligase control of HMGCS1 degradation) — the opposite direction of information flow from what you're proposing. The *synthetic, inducible* degron systems you actually mean (auxin-inducible degron / AID, dTAG) are mature, general-purpose tools, but nobody has used them to force oscillations in mTORC1 specifically, in hepatocytes or otherwise. Two practical notes: tag Raptor, not mTOR itself, to stay mTORC1-specific and spare mTORC2; and degron-mediated degradation + resynthesis is relatively slow, which caps the pulse frequency you can achieve — for fast pulses, chemical pulsed dosing of a bi-steric inhibitor or optogenetic/chemically-induced dimerization would likely outperform a degron system.

degron Raptor AID/dTAG

New hypotheses surfaced by this search

Frequency encoding of mTORC1. ERK, p53, and NF-κB are known to encode information in pulse frequency and amplitude, not average signal. mTORC1 is now known to pulse (cell-cycle oscillation, feeding cycles) but frequency-decoding has never been tested. Experiment: optogenetic or chemically-inducible mTORC1 input + a live biosensor; impose defined pulse trains; read out autophagy (LC3) vs. protein synthesis.

Chronotherapy of mTOR inhibition. Autophagy sensitivity is cell-cycle-phase dependent (Joshi 2024), and mTORC1 is under circadian control (BMAL1/PER2). Does the timing of a rapamycin or bi-steric dose — relative to the clock or the cycle — change efficacy or toxicity? Directly testable, clinically relevant.

Pulsed vs. continuous selective mTORC1 inhibition. Now that mTORC1-selective bi-sterics exist (RMC-6272/RMC-5552), this comparison is possible for the first time. Does intermittent vs. continuous selective dosing differ in triggering autophagy or in provoking resistance? This directly connects the now-resolved Q1 to your original dynamic-activation thesis.

References

Joshi JN, Lerner AD, Scallo F, et al. mTORC1 activity oscillates throughout the cell cycle, promoting mitotic entry and differentially influencing autophagy induction. *Cell Reports* 2024;43(8):114543. doi.org/10.1016/j.celrep.2024.114543

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Sources: PubMed (NCBI) and supplementary web search. All five papers above have been added to the Open mTOR Atlas study base (v27) and the Gaps & Hypotheses tab has been updated accordingly. Generated via the Open mTOR Atlas research pipeline.